Hands-on Guide for Data Engineering

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Agenda

- Introduction to NLP (evolution, core principles)
- Foundational Principles of NLP
- What are Vector Embeddings? Understanding the Foundation of Data Representation
- Text cleaning pipeline
- Hands-on: Text cleaning
- Hands-on: Named Entity Recognition
- Introduction to Large Language Models (LLMs)
- Transformers Architecture

Introduction to NLP

Definition

Natural Language Processing (NLP) is a branch of computer science and artificial intelligence that focuses on the interaction between computers and human language.

Aim

NLP aims to enable computers to understand, interpret, and generate human language in a way that is both meaningful and useful.

Technical Goal

NLP empowers machines to process and analyze vast amounts of textual data, unlocking valuable insights and enabling efficient communication.

Challenges

NLP faces challenges like ambiguity in language, variations in dialects and accents, and the need for robust algorithms to handle complex linguistic structures.

The Evolution of NLP: From Rule-Based Systems to Deep Learning and Transformers

1. ## Rule-Based Systems (1950s-1980s)

Early NLP systems relied on handcrafted rules and linguistic patterns to process language, requiring extensive knowledge of grammar and syntax.

2. ## Statistical NLP (1990s-2010s)

The rise of machine learning brought about statistical NLP, which leverages statistical models trained on large datasets to identify patterns and make predictions about language.

3. ## Deep Learning Era (2010s-Present)

Deep learning, particularly recurrent neural networks (RNNs) and transformers, has revolutionized NLP, enabling models to learn complex language representations and achieve state-of-the-art performance on various tasks.

4. ## Transformers and Pretrained Models (2018-Present)

Introduction of Transformer architecture by Vaswani et al. (2017).
Key Models: BERT (Bidirectional Encoder Representations from Transformers), GPT (Generative Pre-trained Transformer), T5, RoBERTa.
Features: Self-attention mechanism, scalability, and pretraining on massive datasets.
Impact: Significant improvements in a wide range of NLP tasks, state-of-the-art performance.

Unlocking Meaning: Core Concepts in NLP

At its core, NLP centers around breaking down human language into its fundamental components and understanding the relationships between them. Key concepts include:

Text Processing

The foundation of NLP involves cleaning and preparing raw text data, such as removing punctuation, converting to lowercase, and handling special characters, to ensure consistency and improve model performance.

Tokenization

The process of dividing text into individual words or tokens, which serve as the building blocks for further analysis. Tokenization can be as simple as splitting text by spaces or involve more complex rules for handling contractions, punctuation, and special characters.

Stemming & Lemmatization

Techniques for reducing words to their base or root form, such as "running" to "run" or "better" to "good," which helps to consolidate words with similar meanings and reduce vocabulary size, improving model efficiency.

Introduction to NLP. What can be done with NLP

Text summarization

Named entities recognition

Part-of-Speech Tagging
Semantic dependencies

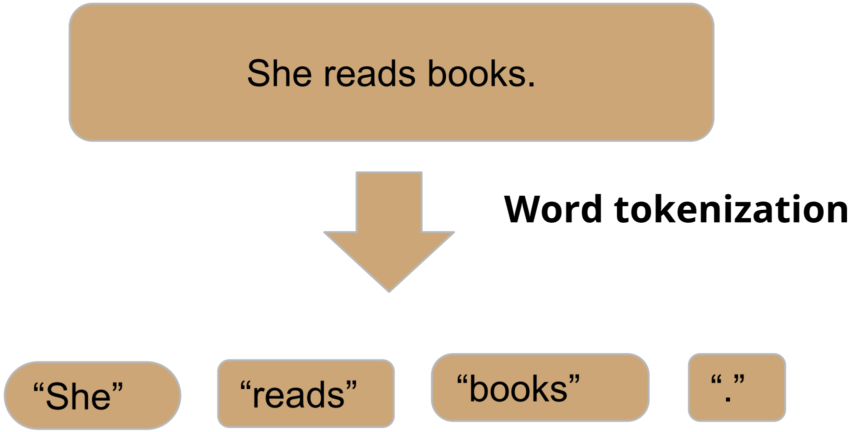
Sentiment Analysis

Text classification

Text Generation

Foundational Principles of NLP - I

Tokenization-first step: splitting text into smaller chunks, typically words or subwords.



Tokenization in Natural Language Processing

Tokenization transforms text into numerical data, crucial for machine learning models in NLP.

The process includes several stages:

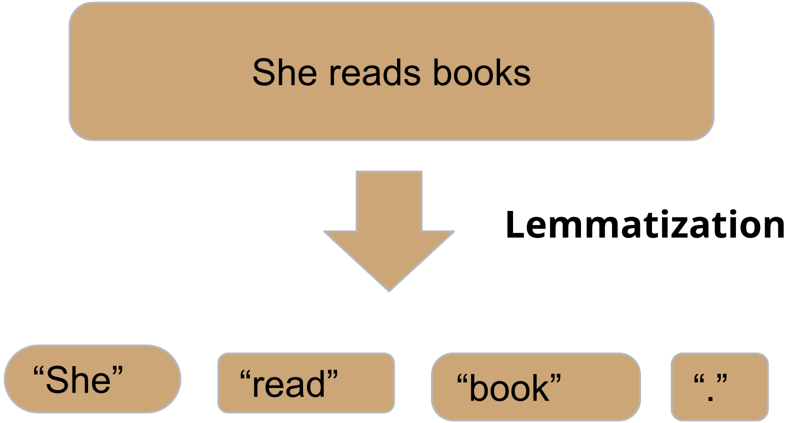
- **Normalization:** Ensures text consistency by cleaning it, like converting letters to lowercase, removing punctuation, or altering accented characters based on the task specifics. While more normalization simplifies data, excessive cleaning can lose context.
- **Pre-tokenization:** Splits the text into segments, such as words, especially when spaces separate them. However, this varies by language.
- **Token Cleanup:** Involves cleaning steps after initial splitting. This might include removing stopwords like "a" or "the," as they add little value to the task at hand.

Balance of Complexity and Context

- **Simple Models:** Require extensive data cleaning to reduce complexity, especially with small datasets.
- **Advanced AI Systems:** Tools like OpenAI's ChatGPT and Google's Gemini retain complexity for nuanced understanding, often leveraging subword tokenization for detailed language analysis.
- **Key Takeaway:** Balance simplicity with essential information to optimize machine learning model training.

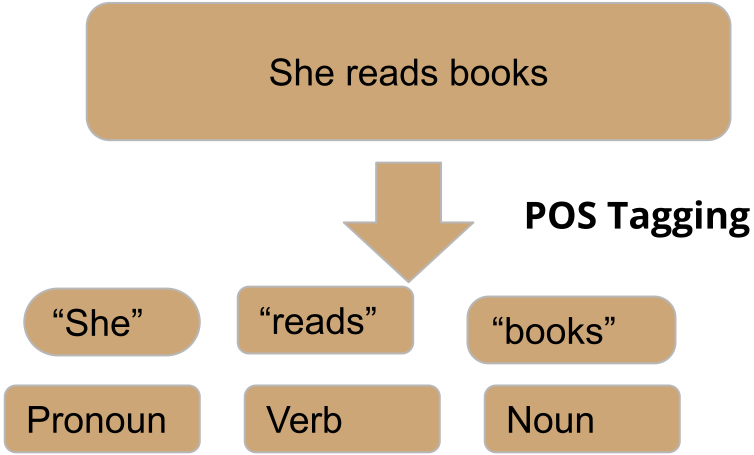
Foundational Principles of NLP - I

Lemmatization: it is a process to change the word to its dictionary form, known as the “lemma”.



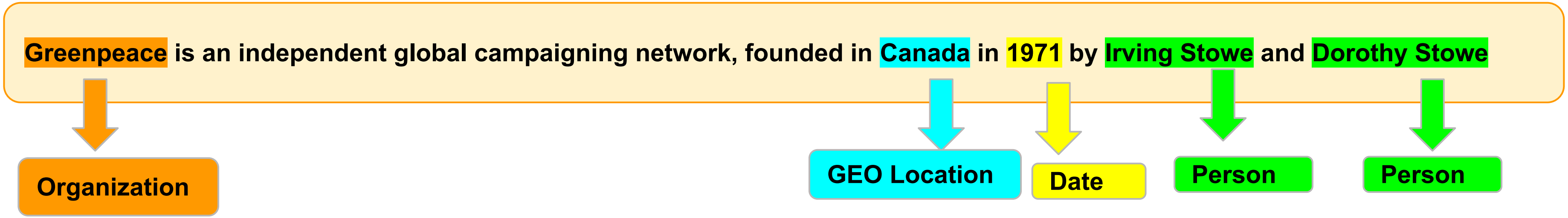
Foundational Principles of NLP - I

Part-of-Speech (POS) Tagging : it is a process of labeling each word (token) in a text with its corresponding part of speech



Foundational Principles of NLP - II

Named Entity Recognition: information extraction from the text that classifies named entities into predefined categories such as the person names, organizations, locations, numbers, specific terms (medical, legal), percentages, etc.



Importance:
Information Retrieval: Helps in improving the accuracy of search systems by focusing on key terms.
Question Answering: Enables systems to answer questions related to specific entities.
Content Recommendation: Helps in content personalization and recommendation by focusing on key entities.
Research: Useful in extracting structured information from massive datasets for academic or corporate research.

Foundational Principles of NLP - II

Sentiment Analysis is a task to determine and extract the sentiment or emotion in the text. The primary goal to detect positive/negative/neutral tone of text.

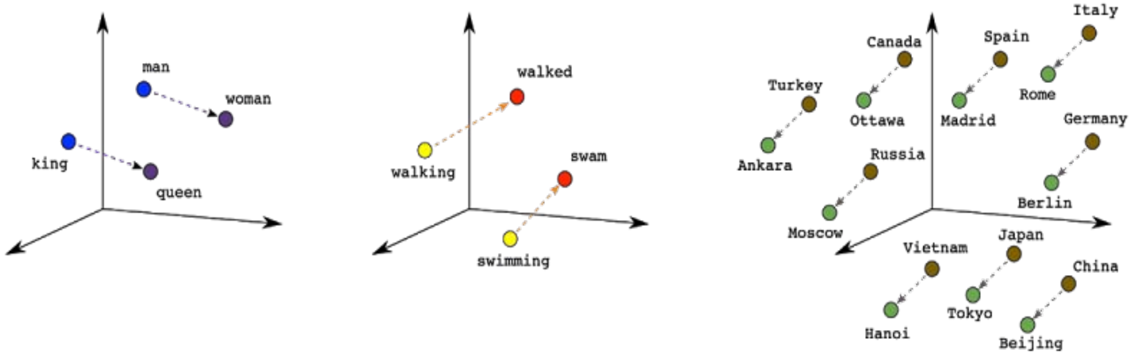
Types:

- **Binary Sentiment Analysis:** Categorizes sentiments as positive or negative.
- **Fine-grained Sentiment Analysis:** Goes beyond binary and might include categories like very positive, positive, neutral, negative, and very negative.
- **Emotion detection:** Determines specific emotions being expressed, e.g., happiness, anger, sadness, etc.

What are Vector Embeddings? Understanding the Foundation of Data Representation

- Embeddings are representations of data as points in space
- Any type of data can be embedded: text, images, videos, users, music
- The key: locations in this space are semantically meaningful
- Think of it as converting data into coordinates with meaning

What are Vector Embeddings? Understanding the Foundation of Data Representation



Male-Female Verb Tense Country-Capital

Embeddings are a way of representing data–almost any kind of data, like text, images, videos, users, music, whatever–as points in space where the locations of those points in space are semantically meaningful.

[Word2Vec]

Some common NLP tools

NLTK (Natural Language Toolkit)

SpaCy (NLP advanced library, industrial standard, supports 60+languages, used for production)

Stanza (formerly StanfordNLP)

TextBlob (simplified text processing)

Gensim, **MonkeyLearn**... and many more

Text cleaning: Complete Text Processing Pipeline. Step 1

Basic Cleaning Steps:

- Remove HTML tags
- Remove URLs and email addresses
- Handle contractions
- Convert text to lowercase
- Remove multiple spaces

Advanced Cleaning Steps:

- Remove special characters, digits, non-ASCII characters
- Correct typos
- Remove emojis
- Perform spelling correction

Text cleaning: Complete Text Processing Pipeline. Step 2

Step 2: Preprocess the Text

- Define custom stop words
- Tokenize, remove stop words, and lemmatize using SpaCy
- Optional stemming using NLTK's PorterStemmer

Hands-on
Text cleaning

Intro to LLMs


Andrej Karpathy
@karpathy
The hottest new programming language is English
12:14 PM · Jan 24, 2023 · 3.8M Views

What exactly is an Large Language Models?

Simple Analogy:

Think of an LLM as a super-smart autocomplete on your phone. It's been fed massive amounts of text and code, and it's learned to predict what might come next.

More Technical:

A "large language model" is essentially a type of computer program adept at understanding and generating human language. It's trained on vast datasets consisting of text and code, enabling it to predict and generate coherent language based on the input it receives.

Large Language Models

A Large Language Model (LLM) is a machine learning model designed to handle and generate human-like text based on vast amounts of training data and trained.
Fondation Models

LLMs in Action: Cool Things They Can Do

1

Natural Language Understanding

LLMs excel at understanding and interpreting human language. They can comprehend context, infer meaning, and provide relevant responses to queries.

2

Text Generation

These models are capable of generating coherent and contextually relevant text. From writing articles to composing poetry, LLMs can produce content in various styles and genres.

3

Language Translation

LLMs facilitate seamless language translation by converting text from one language to another while preserving context and meaning.

4

Code Generation

With their understanding of programming languages, LLMs can assist developers by generating code snippets, automating repetitive tasks, and offering solutions to coding challenges.

5

Creative Collaboration

LLMs can serve as collaborative partners in brainstorming sessions, suggesting ideas, refining concepts, and providing inspiration across diverse fields such as art, design, and storytelling

6

Overall: Increase productivity!

LLMs can help boost productivity by automating tasks, generating content, and providing valuable insights.

“Attention is All You Need.”

by Vaswani et al. in 2017

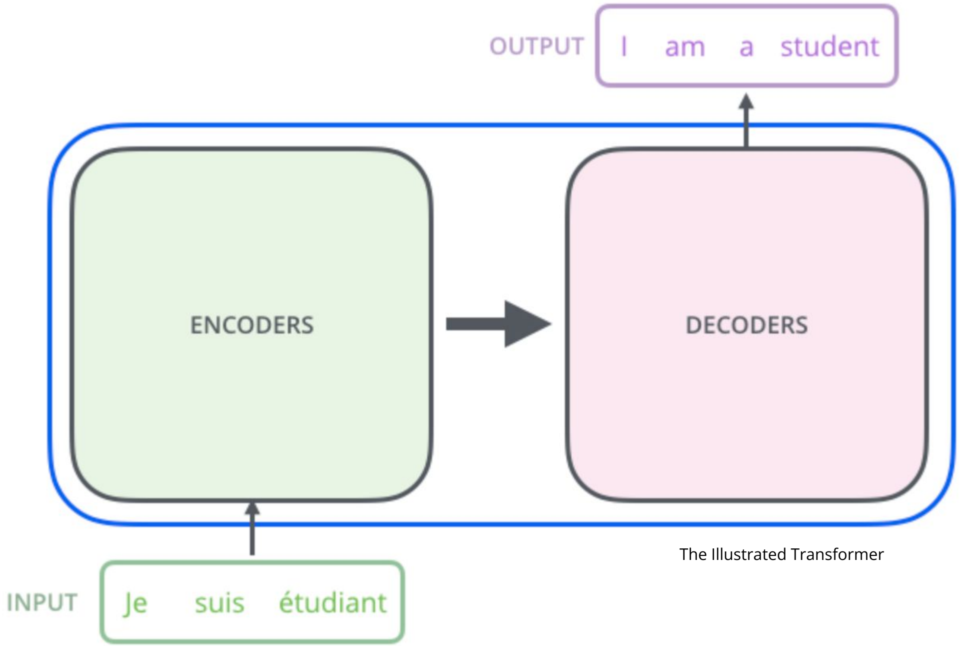
Transformers & Attention Mechanism:

Transformers - new architecture, which relies entirely on self-attention mechanisms.This enables the model to capture relationships in the data regardless of the distance between elements.

She poured water from the pitcher to the cup until it was full.

She poured water from the pitcher to the cup until it was empty.

Transformer architecture



The Illustrated Transformer

Encoder (left): The encoder receives an input and builds a representation of it (its features). This means that the model is optimized to acquire understanding from the input. **The encoder reads and understands text.**
Decoder (right): The decoder uses the encoder's representation (features) along with other inputs to generate a target sequence. This means that the model is optimized for generating outputs. **The decoder writes text based on what it understands.**

Transformers models

Encoder-only models: Good for tasks that require understanding of the input, such as sentence classification and named entity recognition. (BERT (Google), RoBERTa (Meta), Sentence-BERT (SBERT))

Decoder-only models: Good for generative tasks such as text generation. (GPT-family, Llama-family, Google Gemini-family, DeepSeek, Mistral)

Encoder-decoder models or sequence-to-sequence models: Good for generative tasks that require an input, such as translation, summarization, question-answering (Google T5)

Encoder vs Decoder: What Are They Good At?

Encoders (Encode text):

- Semantic search
- Sentiment analysis
- Classification
- Named entity recognition
- Retrieval (for RAG systems)

Decoder-Only Models (LLMs)
Generate text → next-token prediction

- Text generation (chat, stories, explanations)
- Question answering
- Summarization
- Reasoning and chain-of-thought
- Code generation
- Dialogue and interaction
- RAG answer synthesis (not the retrieval part)

Sources to learn more about transformers

How do Transformers work?
https://huggingface.co/learn/nlp-course/chapter1/4

The Illustrated Transformer
http://jalammar.github.io/illustrated-transformer/

Transfer Learning

- Identify the task type for your business problem
- Pick and test a pre-trained model

  -No need to prepare a large dataset

  -Less coding

- Optionally, fine-tune the model on your data

  -Much less data is required

  -No need to train for long periods of time

Large Language Models. Best practices

Achieve best possible performance

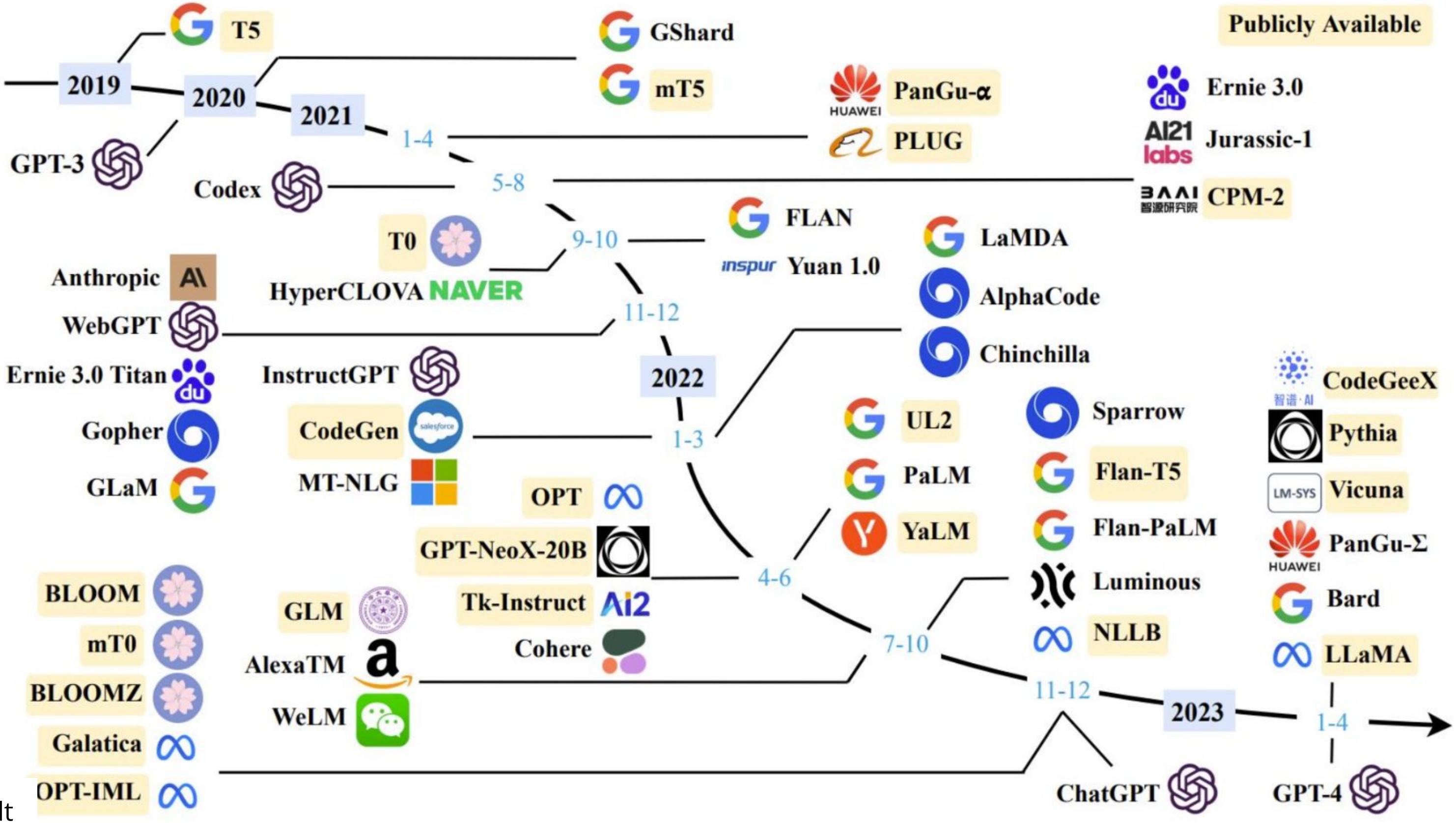
- Use best models (could be LLM or small model depends on the task)
- Use prompts with detailed task context, relevant information, instructions
- Experiment with prompt engineering techniques
- Experiment with few-shot examples that are 1) relevant to the test case, 2) diverse (if appropriate)
- Spend quality time optimizing a pipeline / "chain"

Optimize Cost

- Start optimizing for cost once you reach your expected performance
- Use techniques like Quantization, Distillation or smaller models trained on synthetic data

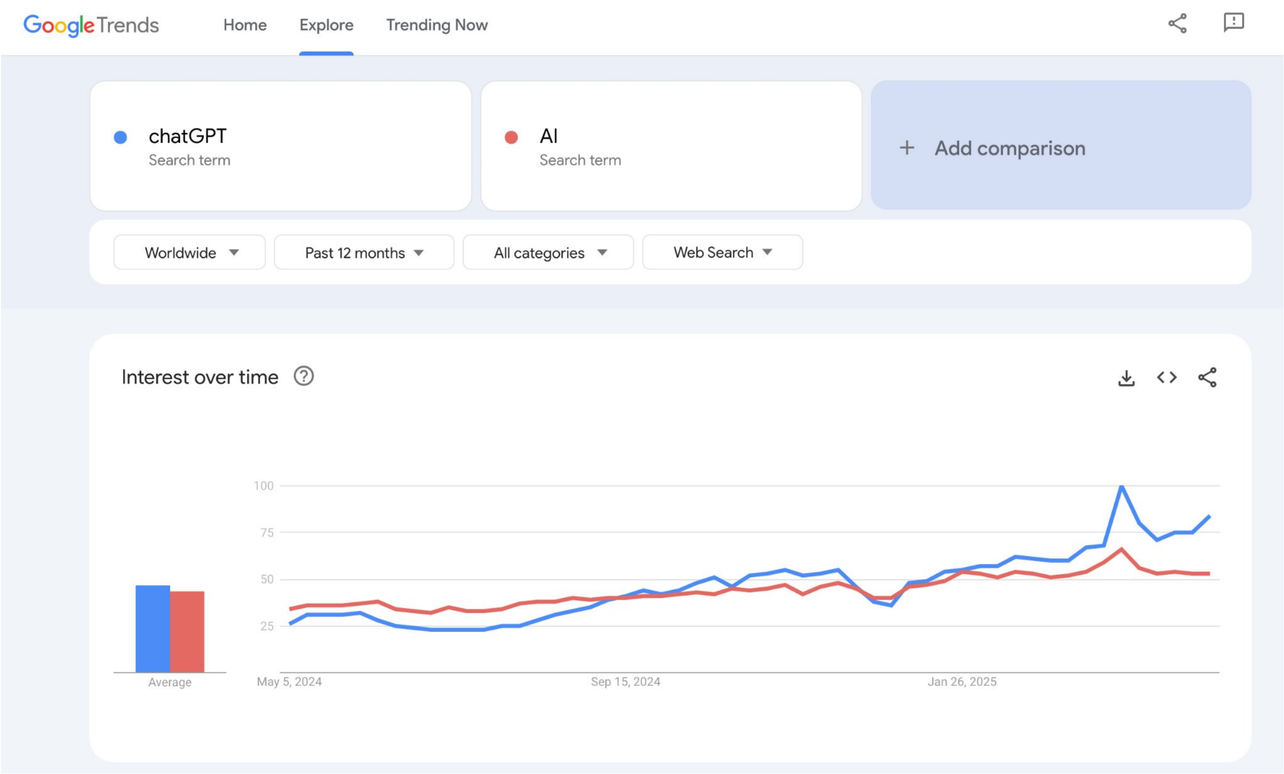
https://arxiv.org/abs/2303.18223

LLMs since GPT-3

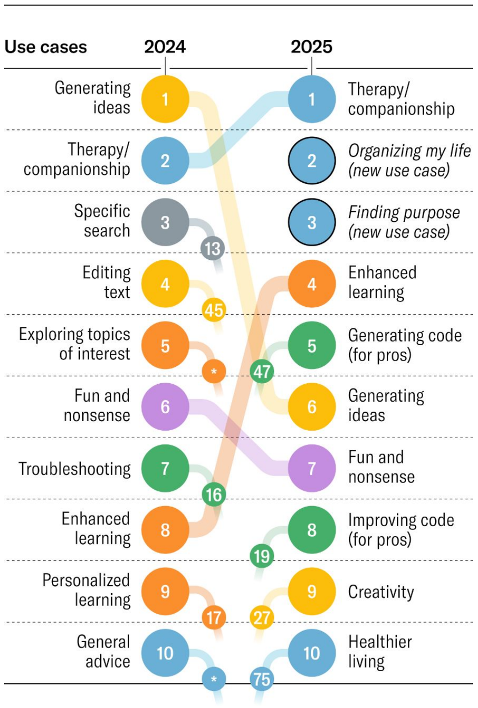


by Philip Schmidt

User interest has doubled in the last 12 months



Top 10 Gen AI Use Cases



Top 10 Gen AI Use Cases

The top 10 gen AI use cases in 2025 indicate a shift from technical to emotional applications, and in particular, growth in areas such as therapy, personal productivity, and personal development.

Themes

- PERSONAL AND PROFESSIONAL SUPPORT
- TECHNICAL ASSISTANCE AND TROUBLESHOOTING
- CONTENT CREATION AND EDITING
- CREATIVITY AND RECREATION
- LEARNING AND EDUCATION
- RESEARCH, ANALYSIS, AND DECISION-MAKING

Harvard Business Review
https://hbr.org/2025/04/how-people-are-really-using-gen-ai-in-2025